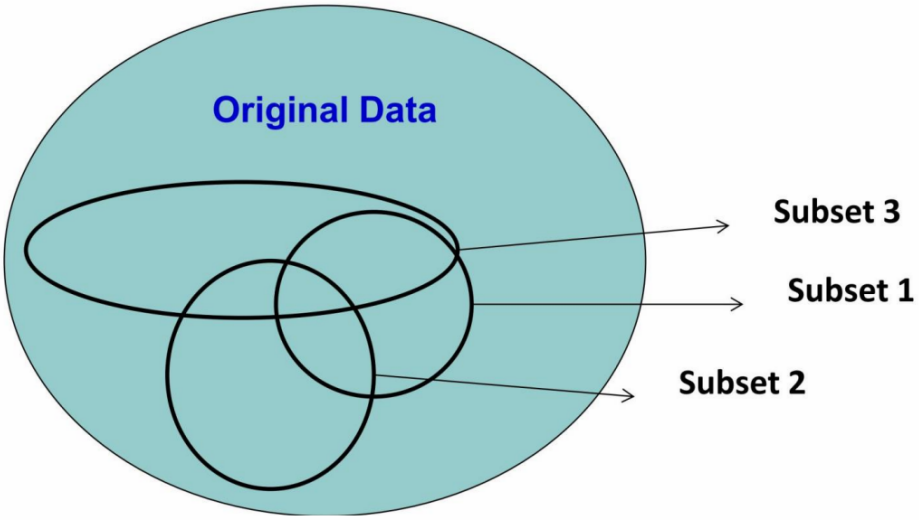
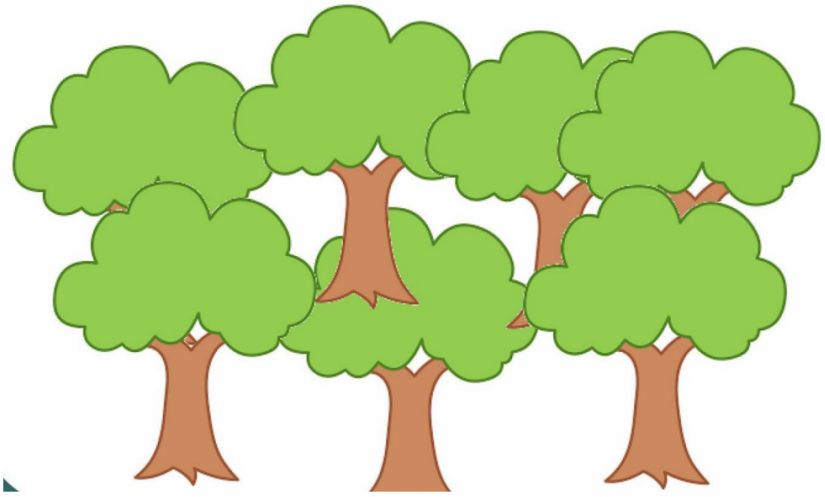
Basic Principals of Random Forest Method

- It develops lots of decision tree based on random selection of data and random selection of variables.
- It provides the class of dependent variable based on many trees.



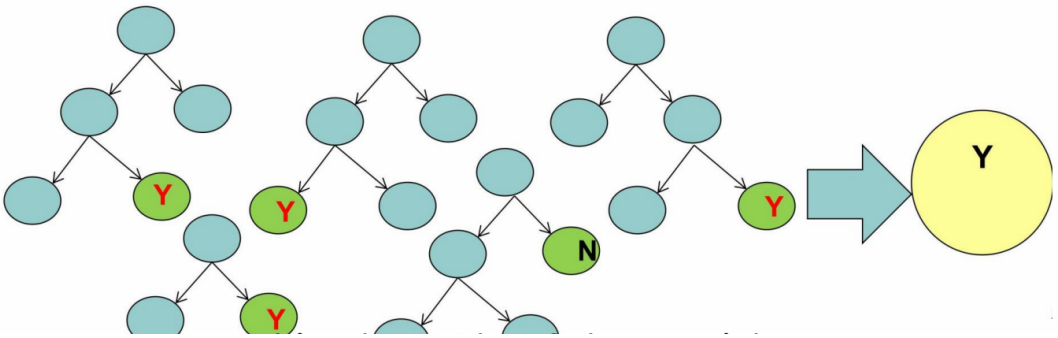
Basic Principals of Random Forest Method

- As the trees are based on random selection of data as well as variables, these are random tree
- Many such random trees leads to a random forest



Basic Principals of Random Forest Method

- Two major belief
  - Most of the tree can provide correct prediction of class for most part of the data.
  - The tree are making mistake at different place
- That's why if we conduct voting for each of the observation and then decide about the class of the observation based on poll result, it is expected to be more close to the correct classification

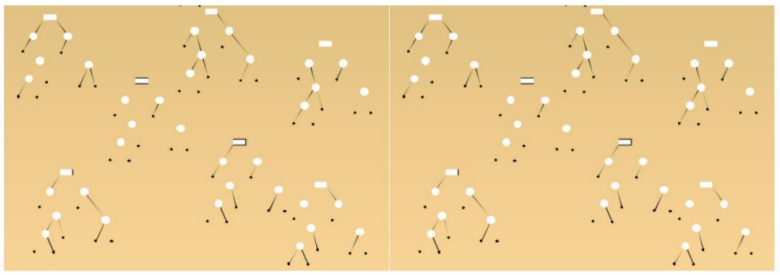


What is Random Forests?

An ensemble classifier using many decision tree models

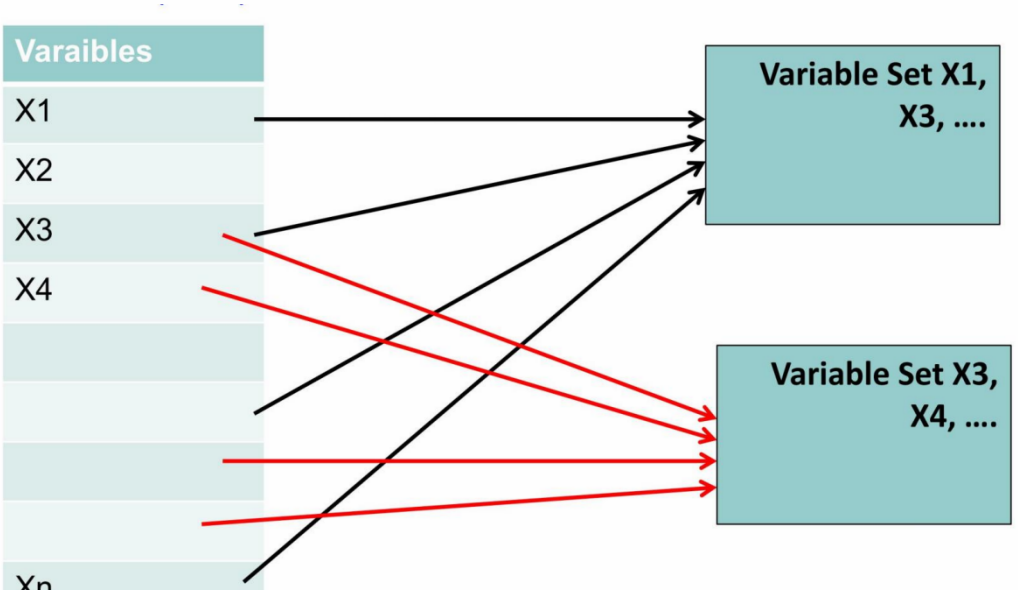
What are ensemble models?

- ✓ Ensemble models combine the results from different models
- ✓ The result from an ensemble model is usually better than the result from one of the individual models



Basic Principals of Random Forest Method

- It develops lots of decision tree based on random selection of data



Features of Random Forests

- ✓ It is unexcelled in accuracy among current algorithms.
- ✓ It runs efficiently on large data bases.
- ✓ It can handle thousands of input variables without variable deletion.
- ✓ It gives estimates of what variables are important in the classification.
- ✓ It generates an internal unbiased estimate of the generalization error as the forest building progresses.



Features of Random Forests

- ✓ It has an effective method for estimating missing data and maintains accuracy when a large proportion of the data are missing.
- ✓ It has methods for balancing error in class population unbalanced data sets.
- ✓ Generated forests can be saved for future use on other data.
- ✓ Prototypes are computed that give information about the relation between the variables and the classification.



The out-of-bag (oob) error estimate

In random forests, there is no need for cross-validation or a separate test set to get an unbiased estimate of the test set error. It is estimated internally, during the run, as follows

- ✓ Each tree is constructed using a different bootstrap sample from the original data.
- ✓ About one-third of the cases are left out of the bootstrap sample and not used in the construction of the kth tree.
- ✓ Put each case left out in the construction of the kth tree down the kth tree to get a classification.
- ✓ In this way, a test set classification is obtained for each case in about one-third of the trees.
- ✓ At the end of the run, take j to be the class that got most of the votes every time case n was oob.
- ✓ The proportion of times that j is not equal to the true class of n averaged over all cases is the oob error estimate. This has proven to be unbiased in many tests.